



PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE
ESCUELA DE INGENIERÍA
DEPARTAMENTO DE INGENIERÍA INDUSTRIAL Y DE SISTEMAS

ICS2523 MICROECONOMICS
COURSE OUTLINE/SYLLABUS

Semester: I'2025
Lecturer: Patricia Galilea (pgalilar@uc.cl)
Course's Schedule: Tuesdays and Thursdays 12:20 to 13:30
Thursdays 14:50 to 16:00
Classroom: K302 on Tuesdays and Thursdays, B22 on Thursdays at 14:50
Credits: 10
Course requirements: ICS1113 Optimización and ICS1513 Introducción a la Economía

I.- INTRODUCTION

Welcome to Microeconomics! Students with low proficiency in English are not advised to take this course. Nevertheless, the main goal of the course is **to learn microeconomics**.

Studying microeconomics will help you understand the world you live in, make you a savvier participant in the economy (you make economic decisions every day), and better understand the potentials and limits of economic policy.

I will do my best to teach it, using different teaching methods in lectures and assignments. Still, I expect an important compromise from you: to take full advantage of this opportunity through **active participation during lectures** (e.g., asking questions, answering my questions, participating in class discussions, etc.), **attending assistantships, studying at home, and doing the assignments**.

We will need the ten credits assigned to this course because we need practice, practice, and more practice to learn English!

II.- LEARNING OUTCOMES

By the end of the semester, students should be able to:

- Apply basic concepts from microeconomics in different contexts, their limitations, and the assumptions on which they are based.
- Identify and differentiate how the market's agents behave, the interaction between these agents, and how it affects the market's performance.
- Apply concepts of introductory microeconomic theory to analyze different mathematical models that explain how consumers and producers interact in a market.
- Understand companies' decision-making process in competitive markets and when market failures exist.
- Understand the concept of efficiency in the allocation of scarce resources.
- **Develop critical thinking through different microeconomic issues using oral and written assignments.**
- **Handle an appropriate microeconomic language in English.**

Each student's participation is necessary to achieve these learning outcomes, requiring a **positive environment**. All members of this course (teachers, teaching assistants, students, and guests) are expected to contribute to a **respectful, welcoming, and inclusive environment for all other class members**. Everyone is welcome to this course, as the learning environment is enriched with diverse people in terms of age, origin, beliefs, ideas, skills, gender, ethnicity, nationality, gender identity, sexual orientation, and many other differences, visible or not visible.

III.- COURSE'S CONTENTS

Although there will be a review of several subjects students might feel familiar with, Microeconomics is more analytical than ICS 1513, and the competencies acquired in ICS 1113 will be helpful. Students are advised to review both courses in detail. The contents of the course are:

1. Introduction (Nicholson, Ch. 1)
 - a. Microeconomic Models
 - b. *Ceteris paribus* assumption
 - c. Optimization in microeconomics
 - d. Positive economy versus normative economy
 - e. Partial equilibrium models versus general equilibrium models
2. Consumer Theory (Nicholson, Ch. 3 to 6)
 - a. Preferences and utility
 - b. Utility functions
 - c. Budget constraint
 - d. Individual consumer problems
 - e. Individual's demand curve
 - f. Indirect utility function
 - g. Expenditure function
 - h. Marshallian and Hicksian demand curves
 - i. Basic identities
 - j. Income and substitution effects
 - k. Slutsky equation
 - l. Consumer surplus and individual welfare
 - m. Aggregated demand
3. Producer's Theory (Nicholson, Ch. 9 to 11)
 - a. Production function
 - b. Returns to scale
 - c. The elasticity of substitution
 - d. Short- and long-run costs
 - e. Producer's behavior
 - f. Producer surplus
 - g. Aggregated supply
4. Welfare Economics and Competitive Markets (Nicholson, Ch. 12 and 13)
 - a. Central planning versus non-central allocation
- b. The price system and competitive market equilibrium
- c. Social welfare functions
- d. General equilibrium versus partial equilibrium model
- e. Pareto efficiency and efficient allocation
- f. Economic efficiency conditions
- g. Efficiency in a competitive market
- h. First and second theorems of economic welfare
5. Market Failure (Nicholson, Ch. 14, 15 and 19)
 - a. Externalities
 - b. Public goods
 - c. Monopoly
 - d. Lerner's rule
 - e. Social welfare analysis under monopoly
 - f. First-degree price discrimination
 - g. Second-degree price discrimination
 - h. Third-degree price discrimination
6. Strategic firm behavior: basic concepts of game theory (Nicholson, Ch. 8 and 18)
 - a. Complete and perfect information games
 - b. Complete and imperfect information games
 - c. Nash equilibrium
 - d. Cooperative oligopolistic markets (cartels)
 - e. Non-cooperative oligopolistic markets (Cournot model, Bertrand model, Stackelberg model)

If you wish to review the lectures at your own pace, you can find 2021's lectures [here](#).

IV.- TEACHING METHODS

The teaching methods included in this course are lectures involving active learning, videos, class discussion, personal reading, essay writing (assignments), video making (assignments), and meme making (assignments). We will use active learning methodologies throughout the course, so **students' involvement in their learning process** (during lectures and at home) **is essential**.

Active learning is a way to engage students in active participation during class. We will use peer-to-peer instruction through several questions that will be asked during lectures. You will have to answer individually first and then with a partner. This method is quite revolutionary, but it has been proven to be an excellent tool for learning, and students tend to like and value it positively.

Since this course is in English, students must be exposed to the English language through videos and special readings. I have selected excellent videos to help introduce the course's contents and make the lecture more enjoyable.

Learning to write and talk effectively is essential, especially since your first years at the Engineering School might have been full of numbers and few opportunities to express your thoughts. But this is very important, not only for the course or to get good grades but for your 'real' life. Alumni who know how to write their ideas clearly and present them to different audiences get better jobs and/or tend to be great entrepreneurs (how do you think they get funding?). You must practice to learn how to write and present well: we will have writing and video assignments throughout the semester. Furthermore, we will also be developing critical thinking.

We will use CANVAS as the course's web page, but we will publish grades in SIDING.

V.- ASSESSMENTS

Dates of **Tests** and **exam**:

Test 1:	Thursday, April 3 rd , 2025 during class hours
Test 2:	Thursday, April 24 th , 2025 during class hours
Test 3:	Thursday, May 22 nd , 2025 during class hours
Test 4:	Thursday, June 5 th , 2025 during class hours
Test 5:	Thursday, June 26 th , 2025 during class hours
EXAM:	Wednesday, July 9 th , 2025 at 08.20 am

The tests and exam may be face-to-face if sanitary conditions allow it.

The Tests' average is called **TA**, and only the best four tests will be considered.

Students must complete **three mandatory assignments**. These assignments are personal¹ and delivered through Canvas. If a student discusses an assignment with someone else, he/she is obliged to state other students' involvement in the fulfilment of this assignment. Also, the student should state any other source of information used (references). It is unacceptable to copy somebody else's assignment or use an old one (even if it's yours). If a student doesn't send the assignment, he/she will have the minimum grade (1,0).

Obligatory assignment deadlines:

Assignment 1:	March 24 th , 2025
Assignment 2:	May 12 th , 2025
Assignment 3:	June 16 th , 2025

The written assignments' average is **AgA**.

Students will have to elaborate on a **meme** (photo only) **or a video** (max 2 min long) presenting and/or explaining in English any subject seen that week or the week before during class. These assignments should be uploaded through Canvas.

	Date	Type of assignment
1	Friday, March 14 th	Meme 1
2	Friday, March 21 st	Video 1
3	Friday, April 11 th	Meme 2
4	Friday, May 9 th	Video 2
5	Wednesday, May 30 th	Meme 3
6	Thursday, June 19 th	Meme 4
7	Friday, June 13 th	Video 3
8	OPTIONAL Thursday, July 3 rd	Video or Meme

¹ Assignments will be sent through the course's web page in CANVAS. No assessment should be sent to the lecturer or teaching assistants.

The meme + video assignments' average is **AgMV**. The last one is optional and may replace one assignment (video or meme, depending on what is delivered).

The rubric for assessing memes and videos will include originality, relevance to the course, correct use of microeconomic terms and concepts, and English proficiency.

The **AgMV** can be calculated according to the following formula:

$$\text{AgMV} = 0.60 * \text{VA} + 0.40 * \text{MA}$$

Where **MA** is the average grades for memes and **VA** is the average grades for videos.

VI.- RULES OF THE GAME

To pass the course, you must fulfill the following requirements:

- a) The Final Grade (FG) must be greater than 3.94

The final grade (FG) can be calculated according to the following formulae:

$$\text{GbE} = 0.60 * \text{TA} + 0.25 * \text{AgA} + 0.15 * \text{AgMV}$$

$$\text{FG} = 0.65 * \text{GbE} + 0.35 * \text{Exam}$$

- b) The AgA should be greater than 3.94. Otherwise, the FG will be the minimum between GbE and 3.9

- If a student misses one test, he or she may erase that grade (the average of the tests only considers the best four grades). If a student misses two tests, one may be erased, and the exam may replace the other grade. If a student misses more than two tests, he/she will have a 1,0 in all the assessments missed except for two, which he/she can replace with the method described above.
- To be exempt from the exam, you must have an **average of the five tests** equal or higher than 5.2.
- Any student doing a dishonest act will have a 1,1 in the course, and the Dirección de Pregrado will be informed of this behavior.
- **Plagiarism** is dishonest and will be punished (in assessments and assignments) and treated as a serious offence. Further description of plagiarism can be found in the following link:

<http://guiastematicas.bibliotecas.uc.cl/normasapa>

- **Use of AI on written assignments:** You may use AI tools for your written assignments as long as you can show your learning in the student outcomes related to developing critical thinking and handling an appropriate microeconomic language in English. Thus, AI tools are restricted to generating ideas for research questions, improving writing quality (but not translating the whole work through AI) and reformatting citations and references.

VI.- POLICY ON CHILDREN DURING CLASS (ADOPTED FROM DR. MELISSA CHEYNEY'S SYLLABUS)

Pontificia Universidad Católica de Chile does not have a formal policy regarding children in the classroom or online. Therefore, this policy only applies to this course and this section for students who are also parents.

1. All exclusively breastfeeding babies are welcome in class as often as necessary to support the breastfeeding relationship. Because pumping milk is not always easy and not all babies reliably take a bottle, I do not want students to feel like they must choose between feeding their baby and continuing their education. You and your nursing baby are welcome in class anytime.
2. For older children and babies, minor illnesses and unforeseen disruptions in childcare often force parents to choose between missing class and staying at home with their child. While this is not meant to be a long-term childcare solution, occasionally bringing a child to class to cover gaps in care is perfectly acceptable.
3. I ask that all students work with me to create a welcoming environment that respects all types of diversity, including diversity in parenting status.
4. When babies and children come to class (points 1 and 2), I ask that you sit close to the door. If your little one needs special attention and is disrupting other students' learning, you may step outside until their need has been met. Non-parents in the class, please reserve seats near the door for your parenting classmates. For the online classes, putting the microphones in silence may help not disrupt others' learning process.
5. I understand that, once you become a parent, often the largest barrier to completing your coursework is the tiredness many parents feel in the evening once children have finally gone to sleep. The struggles of balancing school and childcare are exhausting. I hope that you will feel comfortable disclosing your student-parent status to me. This is the first step in my being able to accommodate any special needs that arise. While I maintain the same high expectations for all students in my classes regardless of parenting status, I am also available to solve any problem with you to make you feel supported as you strive for a school-parenting balance.

VII.- BIBLIOGRAPHY

- 1.- VARIAN, H. (1992), *Microeconomic Analysis*, 3rd Edition, W. W. Norton & Company.
- 2.- NICHOLSON, W. and Snyder, C. (2008), *Microeconomic Theory: Basic Principles and Extensions*, 10th Edition, South-Western Cengage Learning.

Selected readings will be given throughout the semester.